

Clinical Trial Summary

Study NX-447 Phase III synopsis (fictional). Sponsor: Northline Therapeutics. Indication: moderate to severe chronic plaque psoriasis. Design: randomized, double-blind, placebo-controlled, 52-week treatment with optional open-label extension. Primary endpoint: PASI 75 at week 16. Enrollment target was 1,240 participants across 98 sites in nine countries.

Study Population

Inclusion criteria included adults 18-75 years, PASI score 12-40, body surface area involvement at least 10%, and candidate for systemic therapy. Exclusions: prior biologic within 12 weeks, active serious infection, pregnancy, or uncontrolled diabetes with HbA1c above 9%. Mean age was 44.2 years; 41% female; median disease duration 11.3 years.

Intervention

NX-447 administered subcutaneously 150 mg every two weeks after a 300 mg loading dose at weeks 0 and 2. Comparator arms: placebo through week 16 with rescue therapy allowed for severe flare, and active reference adalimumab per label in a secondary stratum (not powered for non-inferiority).

Primary Efficacy Results

PASI 75 at week 16: NX-447 71.4% (n=412) versus placebo 8.1% (n=408), $p < 0.0001$. Absolute difference 63.3% (95% CI 58.1-68.5). PASI 90 achieved in 48.2% of NX-447 patients. sPGA clear or almost clear was 62.7% versus 6.4% placebo.

Secondary Endpoints

Dermatology Life Quality Index improvement ≥ 4 points: 68.9% NX-447 versus 14.2% placebo. Nail psoriasis improvement in modified NAPSI: significant at week 24. Itch numeric rating scale reduction ≥ 4 points at week 4: 54.3% versus 11.0% placebo, supporting rapid symptomatic relief.

Safety Overview

Treatment-emergent adverse events were reported in 62.1% NX-447 versus 58.4% placebo. Serious adverse events: 4.8% versus 5.2%. Discontinuations due to AEs: 3.1% versus 2.7%. No deaths attributed to study drug. Most common events: nasopharyngitis, headache, upper respiratory tract infection, injection site erythema.

Infections and Immunogenicity

Serious infections occurred in 1.2% NX-447 patients (four cases) versus 0.7% placebo. Tuberculosis screening was required at baseline; no active TB cases detected. Anti-drug antibodies developed in 8.4% of NX-447 subjects; neutralizing antibodies in 1.1% with no clear impact on efficacy.

Laboratory Findings

Mild transient neutropenia (<1500 cells/mm³) in 2.3% without clinical sequelae. Liver enzymes elevated above 3x ULN in 1.0% versus 0.5% placebo, all resolved on continued therapy or temporary hold. Lipid shifts were minimal and not clinically significant at week 52.

Subgroup Analyses

Efficacy consistent across prior biologic-exposed and biologic-naïve cohorts. Geographic regions showed PASI 75 rates: North America 73.1%, EU 70.2%, Asia-Pacific 69.5%. Body weight quartiles did not materially modify response.

Open-Label Extension

Week 52 data from extension cohort (n=892): sustained PASI 75 in 78.6%. No new safety signals. Four malignancies reported (basal cell carcinoma, prostate cancer stage I, cervical CIS, melanoma in situ) within expected epidemiological background rates per independent adjudication committee.

Regulatory Status

Rolling submission to FDA accepted Q3 2025. EMA MAA filed with accelerated assessment request. Pediatric investigational plan agreed for ages 12-17 with separate Phase III initiating 2026.

Manufacturing and Supply

Drug substance produced at cGMP facility in Ireland. Cold chain 2-8°C. Shelf life 24 months. Supply agreement covers 36 months post-approval forecast scenarios.

Investigator Conclusions

NX-447 demonstrated clinically meaningful and statistically significant improvement in psoriasis severity with an acceptable benefit-risk profile for the studied population. Long-term registry study NCT-PLACEHOLDER will monitor cardiovascular and malignancy outcomes for ten years.

Patient-Reported Outcomes

Psoriasis Symptom Diary sleep disturbance domain improved by week 2. Work productivity and activity impairment questionnaire showed 31% reduction in absenteeism days at week 16 among employed participants.

Statistical Methods

Primary analysis used Cochran-Mantel-Haenszel test stratified by region and prior biologic use. Multiple imputation for missing data sensitivity analysis confirmed robustness. Non-responder imputation at week 16 yielded conservative estimate still above 65% PASI 75.